An Explorative Study on Abstract Images and Visual Representations Learned from Them

UNIVERSITY OF BIRMINGHAM

BMVC 2025

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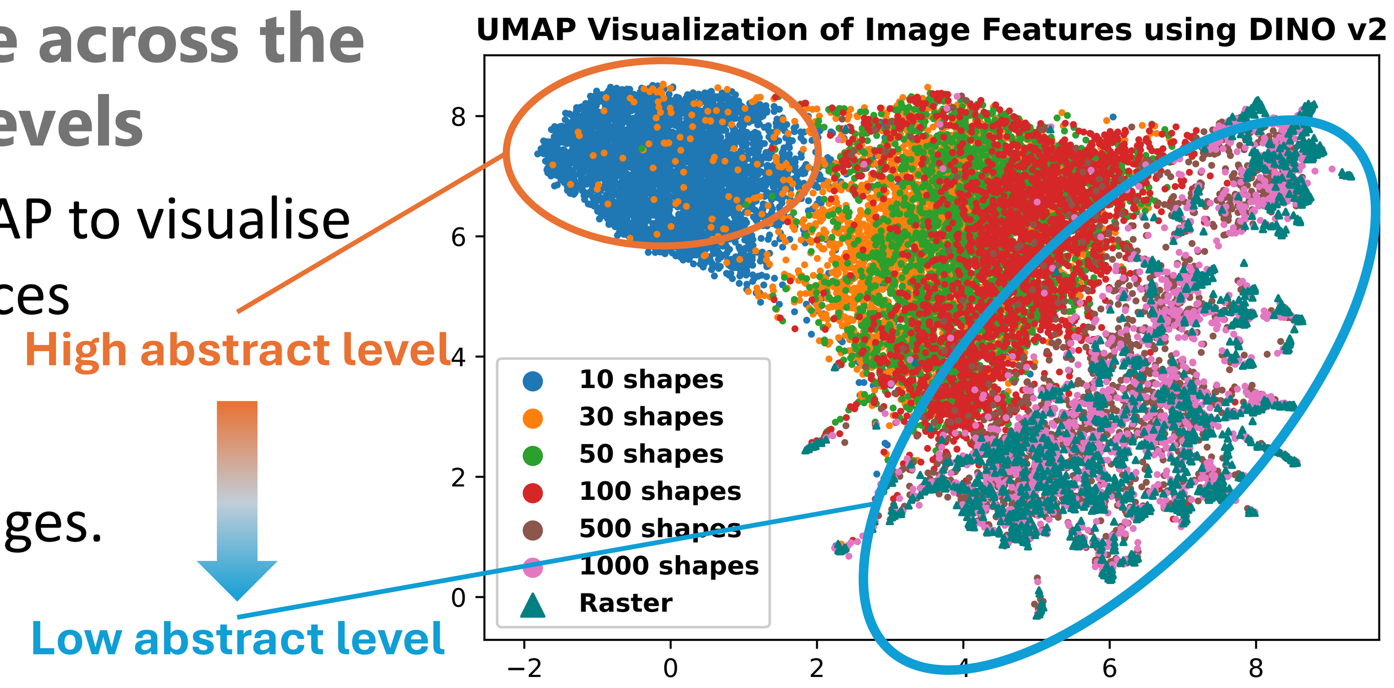
Problems

Abstract images can provide the **Visual Semantics**, however, there is a gap. Why?

- Is **abstraction level** the main reason representations from abstract images perform worse?
- To what extent can abstract images (at various abstraction levels) capture high-level semantics that transfer to **vision downstream tasks**?

Difference across the abstract levels

We use UMAP to visualise the differences between raster and abstract images.



Hierarchical Abstraction Image Dataset (HAID)

Motivation: Current SVG datasets:

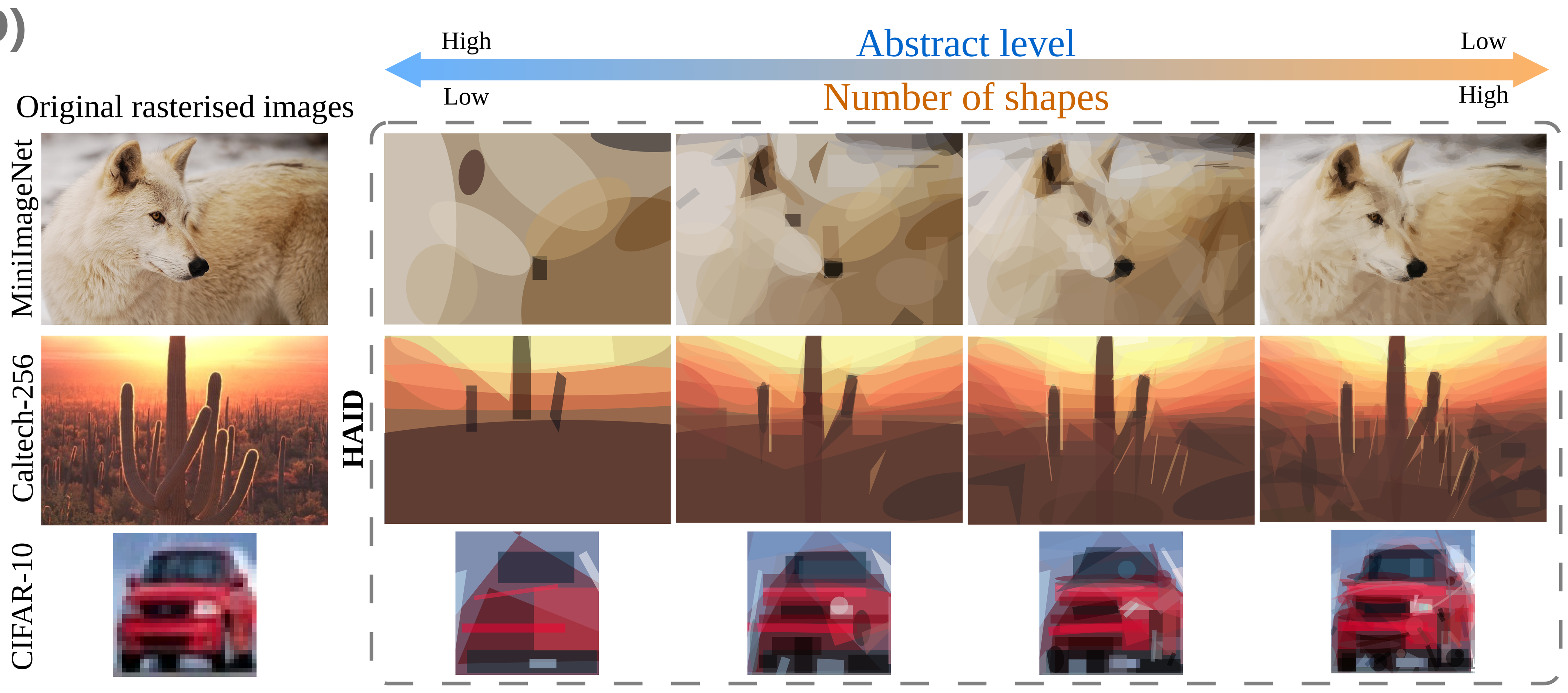
icons emojis fonts

☒ Non-naturalistic

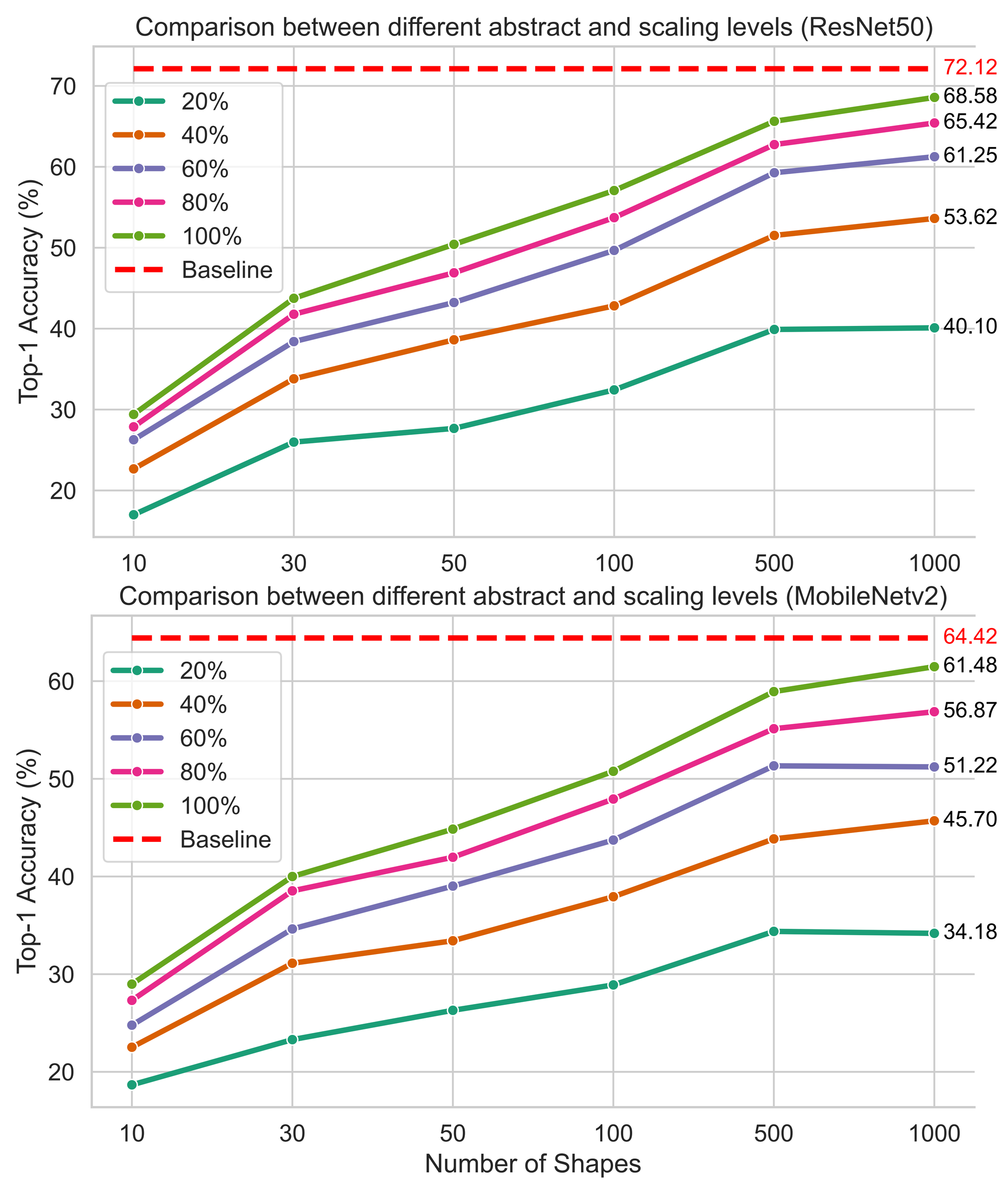
☒ No various abstract levels

Properties of HAID:

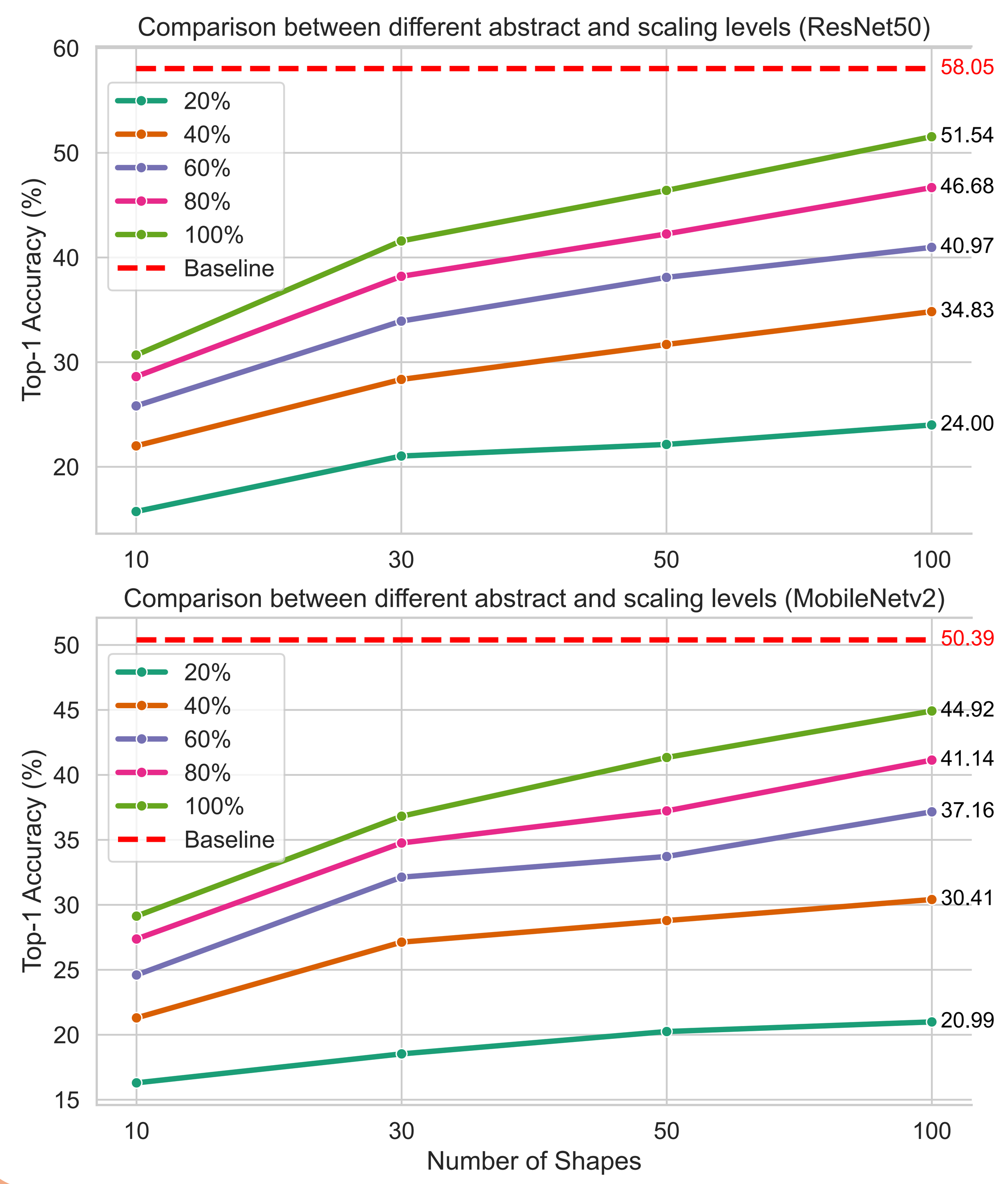
- A **one-to-one correspondence** between the SVG image and their raster image counterparts
- **Multiple abstraction levels** for each raster image.



Classification

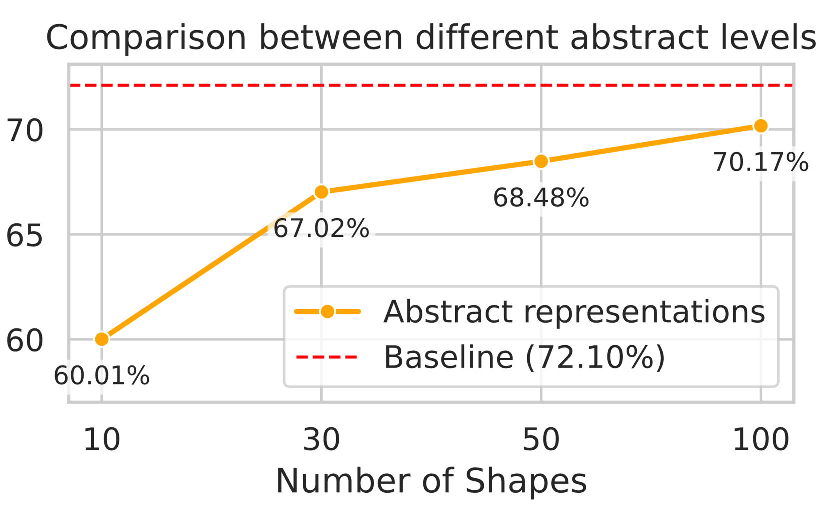


↑ HAID-MiniImageNet

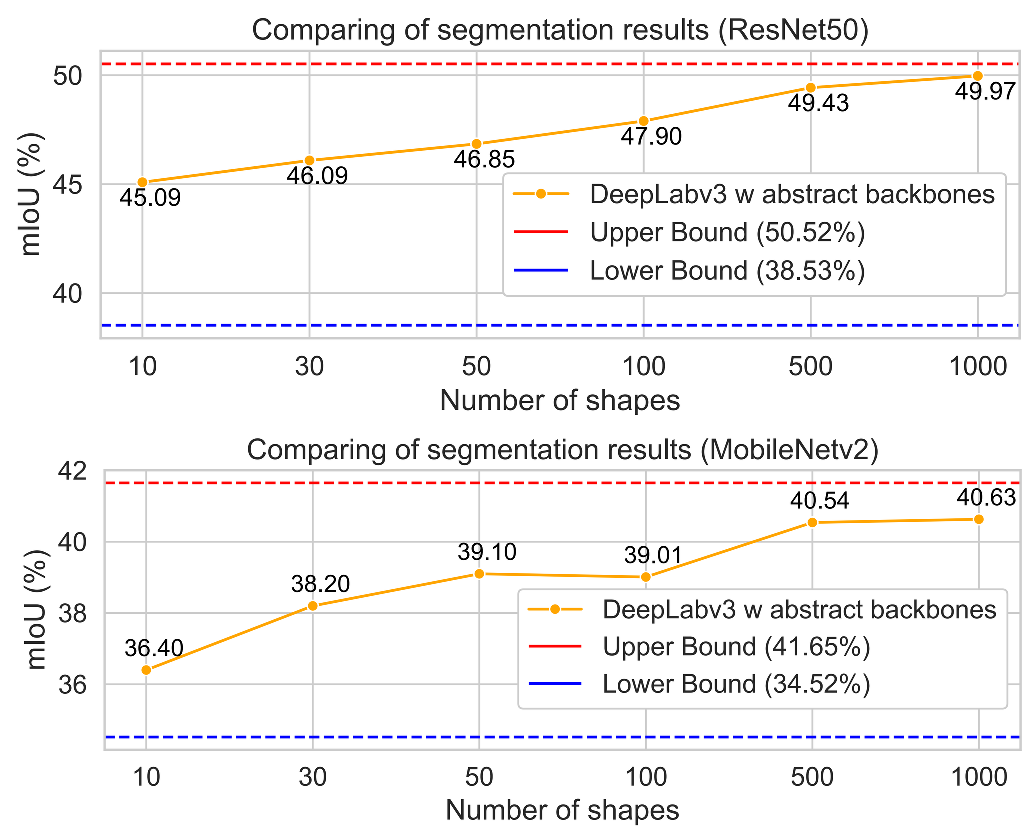


← HAID-Caltect-256

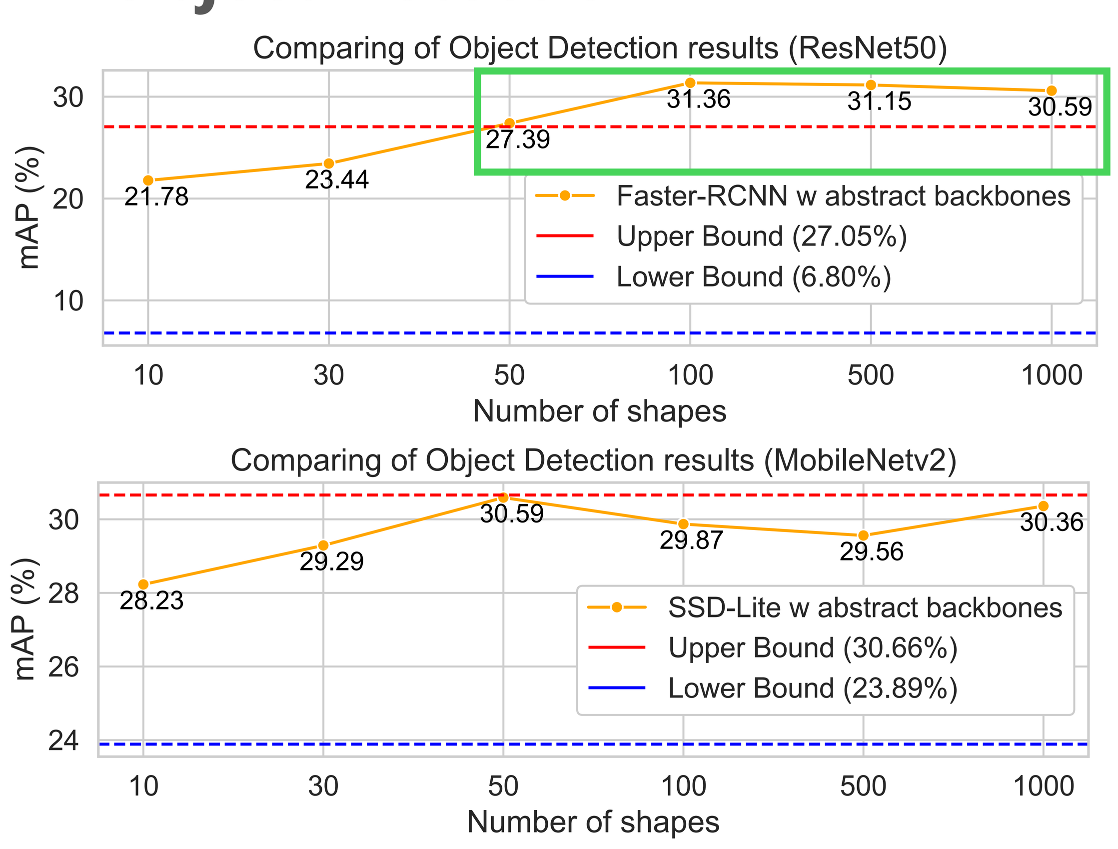
↓ HAID-CIFAR-10



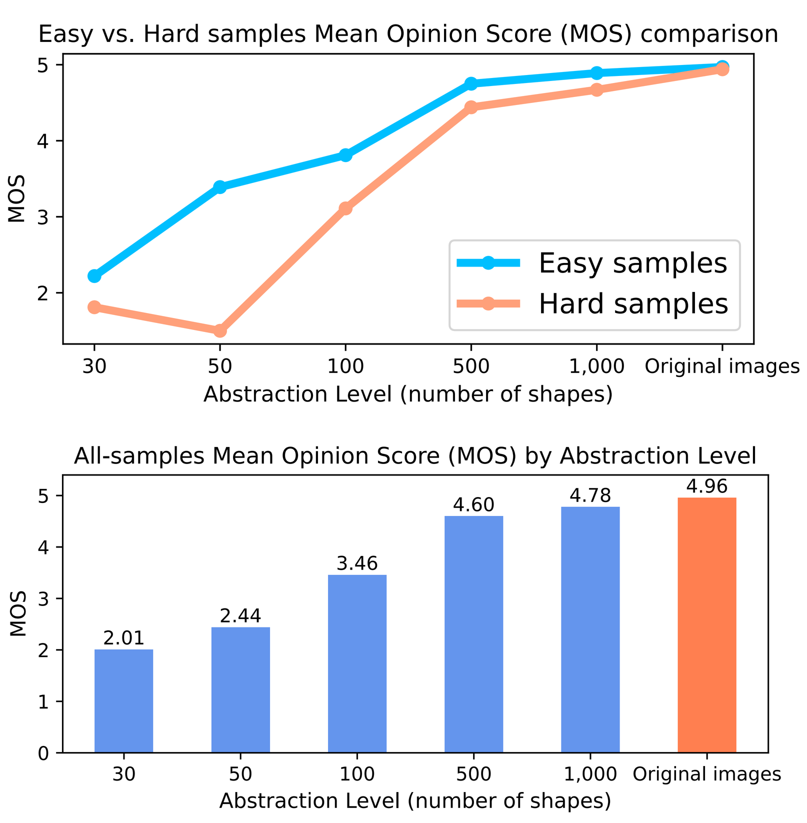
Semantic Segmentation



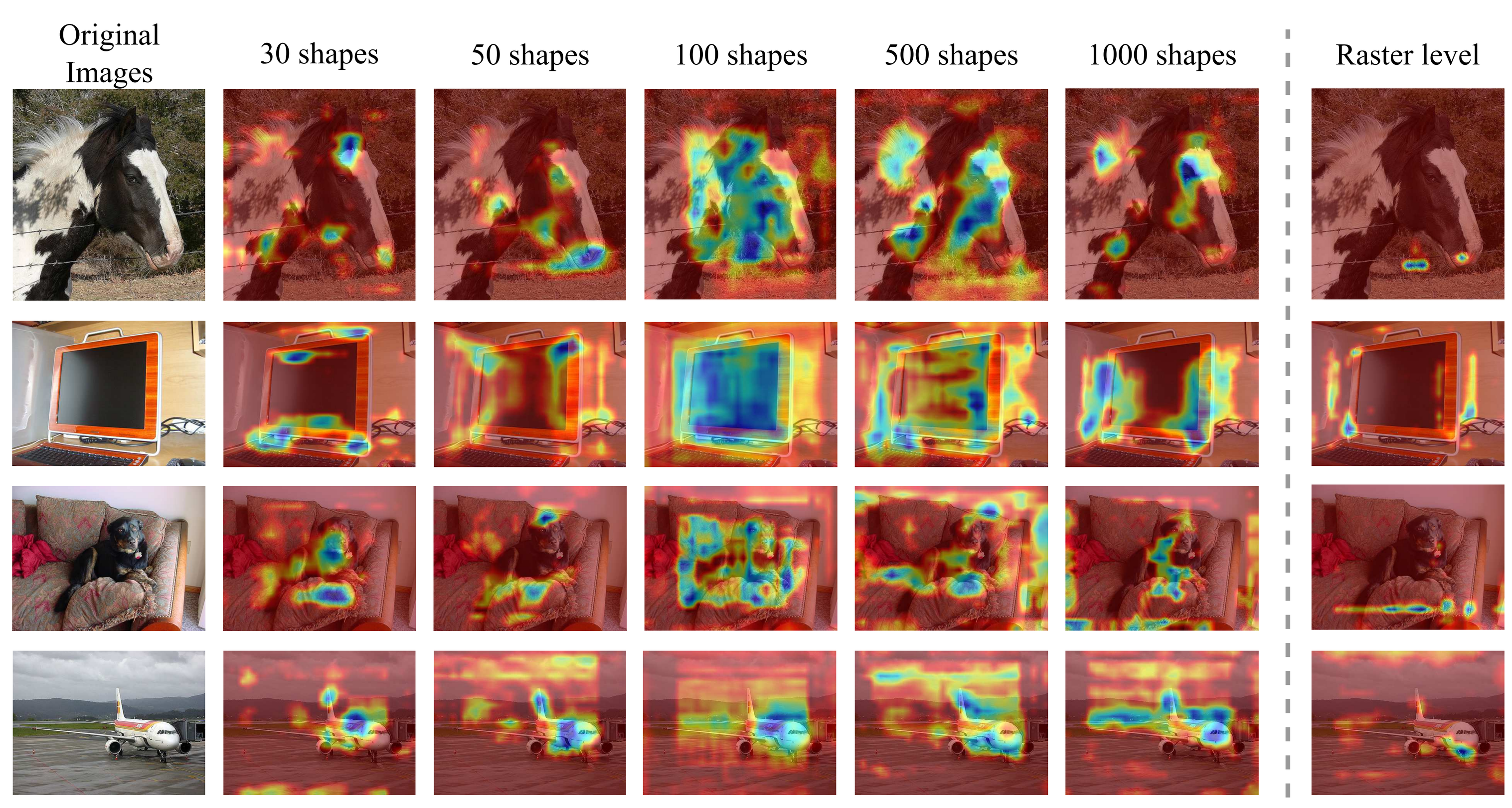
Object Detection



User Study



CAM-visualization results for object detection



Conclusion

- Abstract images can **maintain the visual semantic fidelity** as the **decreasing of the abstract level**.
- Abstract-pretrained backbones **improve localization/detection** and **impose useful geometric priors**.
- Considering their special properties and abilities of maintaining visual semantics, the abstract images are promising to contribute vision tasks.